Lab: Testing

The aim of this lab is to allow you to write a set of tests for your utility functions.

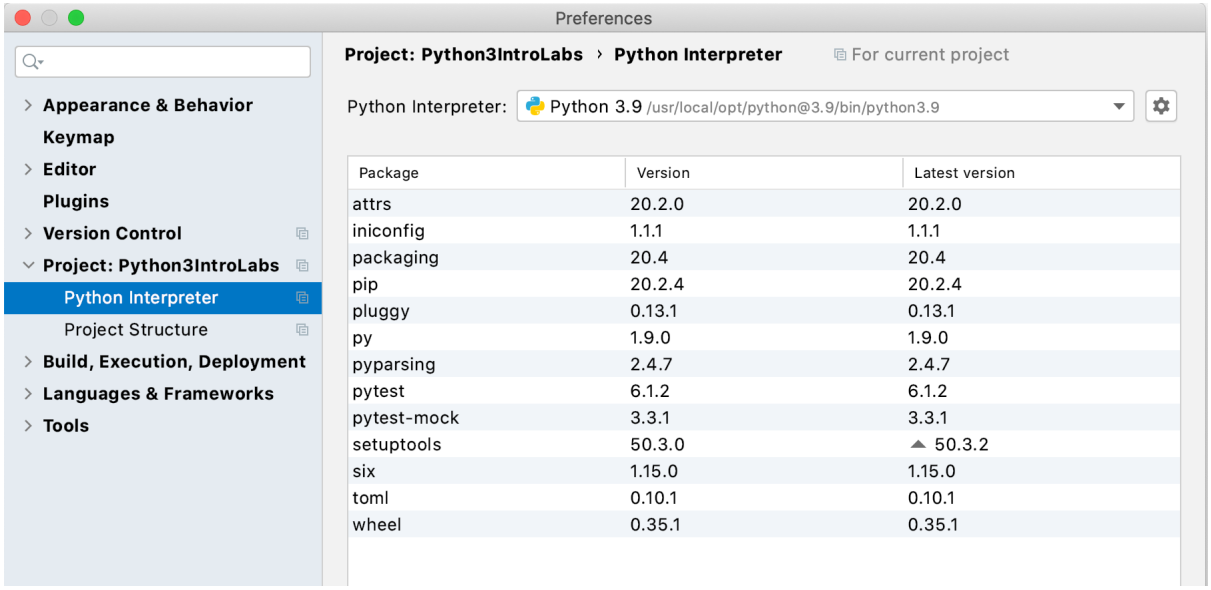
These tests should validate that the functions such as minimum, average, range etc. operate in the way you expect.

Once you have done this now write a set of Tests for your **TemperatureReading** class; what tests do you think you need?

Don't forget to add pytest to the list of libraries on your pythonpath for your Python environment.

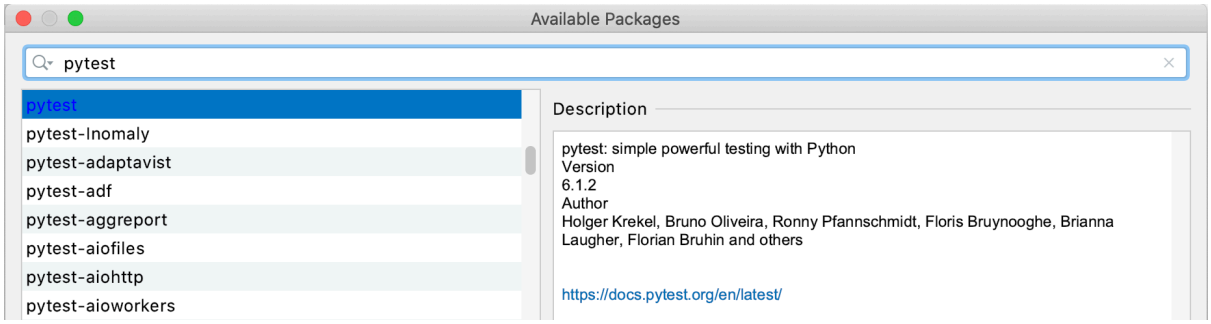
This can be done interactively through PyCharm.

Open your Preferences (on a Mac) Settings (on Windows) and select 'Project: <your proj name>' and then the 'Python Interpreter':



You can then add a package to the current environment for the specified interpreter using the '+' button at the bottom of the dialog.

To add pytest type in pytest to the resulting search bar and click install:



Note working with floating point numbers is notoriously difficult when running tests.

To help with this Python provides the `isclose()` function from the `math` module.

This must be imported into your test file, for example:

```
from math import isclose
```

You can then use this with an assertion, for example:

```
def test_average_int():
   result = average(data)
   assert isclose(result, 4.1, rel_tol=0.05)
```

The rel_tol indicates the relative tolerance – it is the maximum allowed difference between the two values being compared. 0.05 indicates a tolerance of 5%.

The following provides a simple test file for the utils module:

```
from math import isclose

from utils import *

data = [
      2, 3, 4, 5, 6, 5, 4
]

def test_min_int():
   result = minimum(data)
   assert result == 2

def test_max_int():
   result = maximum(data)
   assert result == 6

def test_average_int():
   result = average(data)
   assert isclose(result, 4.1, rel_tol=0.05)
```

As an extension point now write versions of the above for your temperature classes.